

Pothole Detection using Deep Learning: A Comparative Study of YOLO Models.

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Abstract: Potholes pose a significant risk to road safety and vehicle integrity, particularly on rural and high-traffic roads. This paper presents a comparative analysis of YOLO (You Only Look Once) object detection models, specifically YOLOv8s and YOLOv11s, for pothole detection using computer vision. The study utilized a diverse dataset consisting of annotated images of potholes with varying shapes and sizes, collected under standard daylight conditions. The annotations distinguished between freshly formed and well-established potholes to improve detection precision. Results showed that YOLOv8s achieved higher detection accuracy and precision (0.7501), making it well suited for automated road maintenance systems. In contrast, YOLOv11s demonstrated faster inference speeds, favoring its application in real-time scenarios on resource-constrained devices. This study highlights the potential of deep learning-based object detection to enhance road safety and infrastructure monitoring through more efficient pothole identification.

Keywords: Pothole Detection, Yolo Only Look Once, Deep Learning, Computer Vision, Object Detection.

1 INTRODUCTION

Potholes represent a major hazard to road users, often leading to vehicle damage, accidents, and traffic disruptions. Timely identification and repair of potholes are critical for ensuring road safety and minimizing maintenance costs. Traditional detection methods, which rely heavily on manual inspection and public reporting, are time-consuming, labour-intensive, and prone to delays. With the rapid advancement of deep learning and computer vision, automated pothole detection systems have emerged as a viable solution. These systems leverage object detection models to identify potholes in real-time from road surface imagery, enabling faster response and preventive maintenance. Among the state-of-the-art object detection algorithms, the YOLO (You Only Look Once) family stands out due to its speed, accuracy, and efficiency in handling real-time detection tasks. In this study, we evaluate the effectiveness of two YOLO models—YOLOv8 and YOLOv11—for pothole detection. The dataset used for training and evaluation is sourced from Roboflow and contains annotated images of potholes of various sizes and shapes. Annotations are provided in the form of bounding boxes, allowing the models to learn to localize and classify potholes accurately. The primary objective of this research is to compare the performance of YOLOv8 and YOLOv11 in terms of detection accuracy, robustness, and inference speed. Performance is assessed using standard metrics including Precision, Recall, Mean Average Precision (mAP), and Intersection over Union (IoU). The goal is to determine which model offers better performance for pothole detection tasks under general conditions. The methodology includes preprocessing the Roboflow dataset, training both YOLO models, and conducting a detailed performance analysis. The results aim to provide insights into selecting an optimal YOLO model for real-time pothole detection systems that can be integrated into road monitoring or maintenance applications.

2 LITERATURE REVIEW

Bhavana N et al. [1] proposed a real-time pothole detection system called POT-YOLO, integrating YOLOv8 with edge segmentation techniques. Road videos were converted into image frames, pre-processed using a Contrast Stretching Adaptive Gaussian Star Filter (CAGF), and processed via a Sobel edge detector before detection. The method achieved a detection accuracy of 99.10%, with a precision of 97.6%, recall of 93.52%, and F1-score of 90.2%, outperforming other models on benchmark datasets. Surya Sasank Ch and Teja Tallam [2] presented an automated pothole detection and size estimation system using a

combination of CNN and YOLO models. Trained over 200 epochs, the system achieved a mAP@0.5 of 92% on a custom dataset and incorporated dimension estimation based on spatial resolution factors extracted from images, offering a comprehensive non-destructive inspection tool. Arjun Paramarthalingam et al. [3] developed a YOLO-based solution tailored to assist visually impaired individuals in detecting potholes in real-time. Incorporated into a smartphone application, the system provides auditory or haptic feedback, achieving an accuracy of 82.7% and a frame rate of 30 fps for live video streams. Although highly beneficial for enhancing pedestrian safety, the solution is currently limited to detecting potholes in close proximity. Malhar Khan et al. [4] addressed pothole detection in the context of autonomous vehicles by developing a YOLOv8-based deep learning model. The proposed system achieved perfect accuracy and F1 score (both 1.0), surpassing previous YOLO versions in precision and speed, providing robust pothole detection capabilities for self-driving vehicle safety systems. Mostafa Hoseini et al. [5] developed a low-cost, dashcam-based pothole detection system for forest roads using YOLOv5 integrated with the StrongSORT tracking algorithm. Achieving a precision of 0.79, recall of 0.58, and mAP@0.5 of 0.70, the system demonstrated the viability of deploying consumer-grade dashcams and deep learning models for rural and remote road monitoring. Yashar Safyari et al. [6] provided a comprehensive review of pothole detection methods, focusing on computer vision and machine learning techniques. The paper emphasizes hybrid approaches combining traditional image processing with machine learning, which deliver higher accuracy in pothole detection. The review advocates for the use of deep learning, especially CNNs, for improved detection rates, while also highlighting the continued relevance of traditional methods. The paper discusses various sensing systems, algorithms, and future research directions for effective automated road surface assessments. Lincy et al. [7] developed a pothole detection system using YOLOv7, achieving real-time detection with 94.5% accuracy. The system alerts drivers through buzzer warnings, significantly improving road safety. In addition to potholes, the system is being expanded to detect other road irregularities like humps, offering superior performance compared to previous methods. The research contributes to advancing driver assistance systems and minimizing accidents caused by road damage.

3 METHODOLOGY

3.1 Architectural Diagram

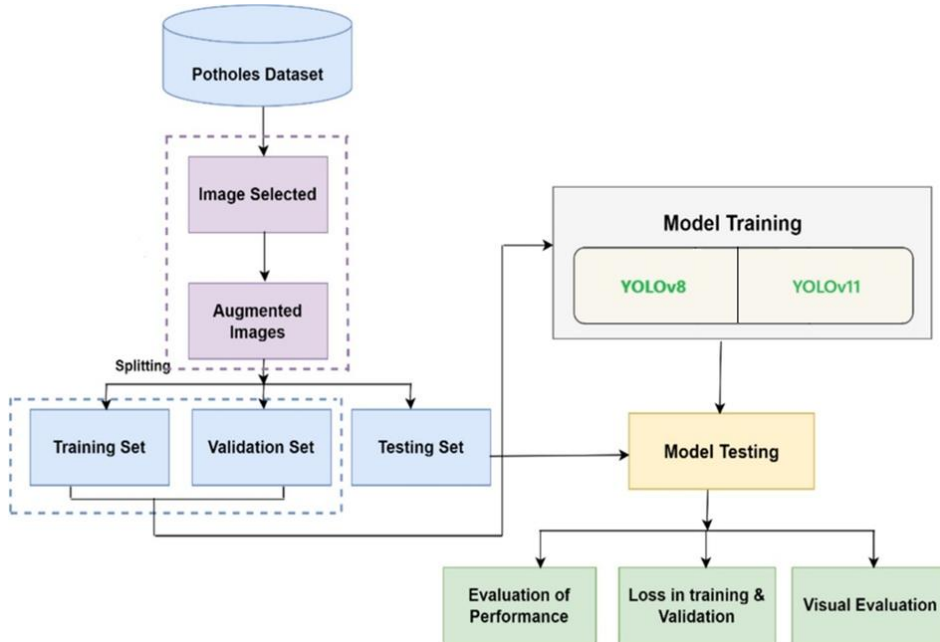


Fig 1: Architectural diagram of Pothole detection

Figure 1 shows the methodology, the process begins with a pothole dataset, which includes annotated images of road surfaces containing potholes. These images undergo data augmentation, including transformations such as tilting, flipping, and colour adjustments to enhance the model's robustness. The augmented

dataset is then split into training, validation, and testing sets, where 80% of the images are used for training, 10% for validation, and 10% for testing. The model training phase utilizes two versions of the YOLO (You Only Look Once) algorithm—YOLOv8 and YOLOv11—to detect potholes efficiently. Both models are trained separately, and their performances are compared to select the best-performing model. The training process involves convolutional layers that extract important features, followed by batch normalization and dropout techniques to prevent overfitting. Once training is complete, the model testing phase assesses the trained models using the test dataset. This phase evaluates performance based on key metrics such as precision, recall, mean average precision (mAP@0.5, mAP@0.5:0.95), and overall detection accuracy. Additionally, the loss during training and validation is monitored to ensure model stability and convergence. Finally, a comprehensive evaluation is performed through three key methods Firstly Performance Metrics Analysis – Measuring accuracy, recall, precision, and F1-score Second Loss Monitoring – Tracking loss trends during training and validation to ensure optimal learning and finally Visual Evaluation – Comparing detection results with ground truth annotations to analyze false positives and false negatives.

3.2 Datasets

The dataset for this paper is obtained from Roboflow Universe[8] and contains annotated road scenes specifically designed for pothole detection, aiding in training object detection models. The Roboflow Pothole Detection Dataset includes a total of 3,393 images with potholes as the only class. The dataset follows the YOLO annotation format with bounding box annotations and is split into 80% training, 10% validation, and 10% testing. Preprocessing steps include auto-orientation, resizing images to 640×640 pixels, and filtering null images to ensure all contain annotations. Various augmentation techniques are applied, generating three outputs per training example. These techniques include horizontal and vertical flips, 90° rotations (clockwise, counter-clockwise, upside-down), random rotations between -15° and +15°, hue adjustments (-20° to +20°), saturation variations (-30% to +30%), exposure changes (-12% to +12%), blur (up to 1px), noise (up to 1.95% of pixels), and mosaic augmentation. Additionally, bounding boxes undergo transformations such as horizontal and vertical flips, rotations between -10° and +10°, and noise adjustments (up to 0.1% of pixels). The images are scaled and normalized to enhance model learning.

3.3 Technology and Frameworks used

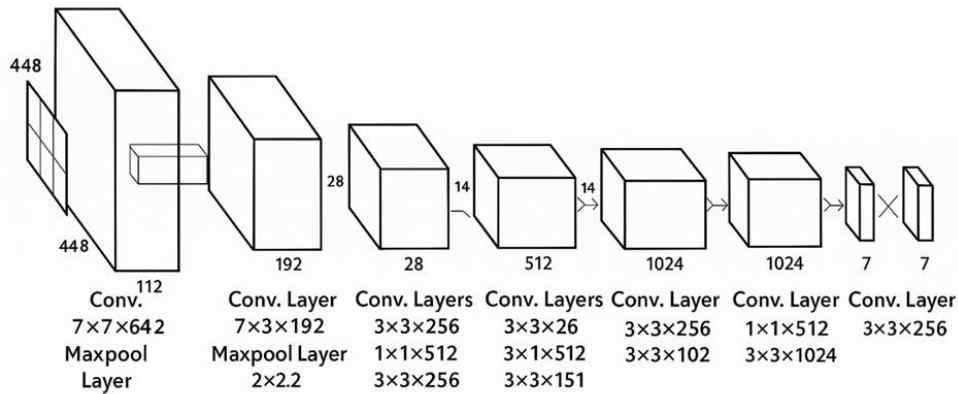


Fig 2: YOLO architectural diagram

Figure 2 shows the architecture processes input images by first resizing them to 448×448 before passing them through the convolutional network. A 1×1 convolution is applied to reduce the number of channels, followed by a 3×3 convolution to generate a cuboidal output. The model primarily utilizes the Rectified Linear Unit (ReLU) activation function, except in the final layer, which employs a linear activation function. Additionally, techniques such as batch normalization and dropout are incorporated to regularize the model and prevent overfitting, ensuring better generalization and stability during training. The pothole detection system was implemented using Python with the PyTorch deep learning framework for training YOLO models on custom datasets, and Google Colab as the development environment providing GPU support for efficient model training and evaluation.

4 RESULTS

4.1 Comparison of YOLOv8 Models

Metric	YOLOv8n	YOLOv8s	YOLOv8m	YOLOv8l	YOLOv8x
Precision	0.6570	0.7501	0.6723	0.6203	0.6570
Recall	0.6065	0.5786	0.5842	0.6036	0.6065
mAP@0.5	0.6425	0.6887	0.6437	0.6526	0.6425
mAP@0.5:0.95	0.3052	0.3314	0.3155	0.3026	0.3052

Table 1: YOLOv8 Model Evaluation Results

Table 1 illustrates that YOLOv8s was chosen from the YOLOv8 series due to its superior precision (0.7501) and mAP@0.5 (0.6887), making it the most effective model for reducing false positives and enhancing detection accuracy. Additionally, its mAP@0.5:0.95 score of 0.3314 indicates consistent performance across varying overlap thresholds, further establishing it as the optimal choice for high-accuracy pothole detection applications.

4.2 Comparison of YOLOv11 Models

Metric	YOLO11n	YOLO11s	YOLO11m	YOLO11l
Precision	0.6575	0.6981	0.6602	0.7118
Recall	0.6219	0.6267	0.6333	0.6075
mAP@0.5	0.6509	0.6733	0.6579	0.6628
mAP@0.5:0.95	0.3196	0.3320	0.3144	0.3154

Table 2: YOLOv11 Model Evaluation Results

Table 2 shows YOLO11s was chosen from the YOLOv11 series because of its balanced performance. While it has a slightly lower precision (0.6981) compared to YOLOv8s, its recall (0.6267) is higher, ensuring fewer potholes are missed. It also achieved a competitive mAP@0.5:0.95 of 0.3320, which further supports its robustness. YOLO11s is better suited for real-time applications where a balance between accuracy and inference speed is critical.

4.3 Confusion Matrix for YOLO8s & YOLO11s

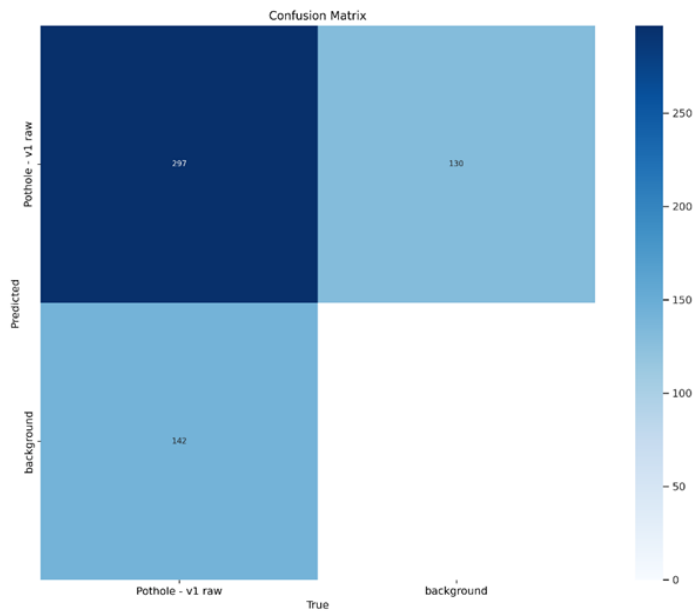


Fig 3: Confusion matrix of YOLOv8s

The confusion matrix (Fig 3) shows that the YOLOv8s model detected 297 potholes correctly, with 130 false positives and 142 true negatives. This indicates strong detection ability, though further improvement is needed in minimizing false alerts.

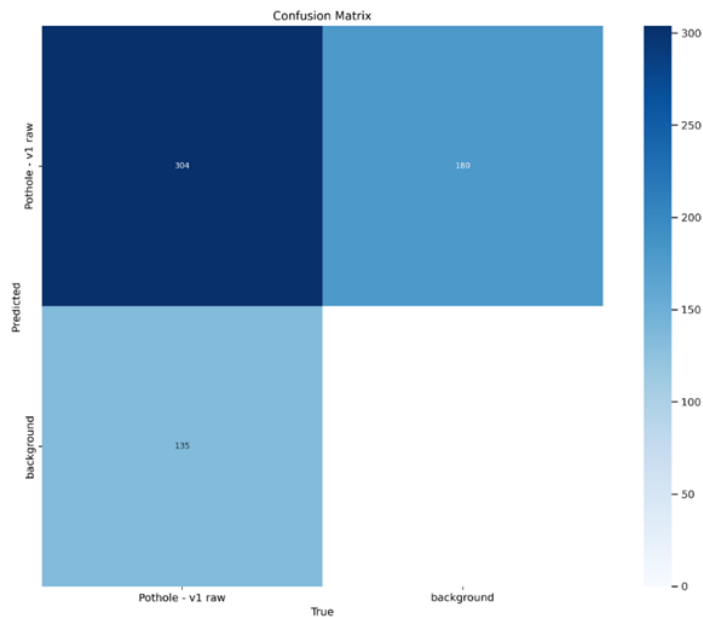


Fig 4: Confusion matrix of YOLOv11s

The confusion matrix (Fig 4) for the YOLO11s model shows 304 correct pothole detections, with 180 false positives and 135 true negatives. This reflects strong detection performance, though with some increase in false alerts compared to YOLOv8s.

4.4 Training Loss Graph for YOLOv8s & YOLOv11s

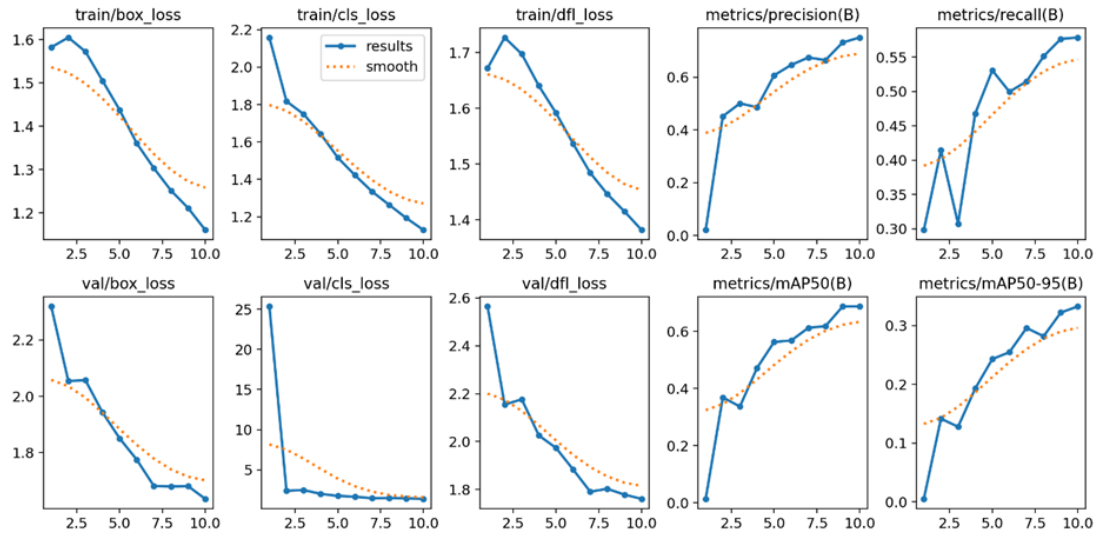


Fig 5: Training Loss Graph for YOLOv8s

Figure 5 shows the training loss curve for the YOLOv8s model, which gradually decreases over time with minor fluctuations in later epochs. This indicates steady learning, though further hyperparameter tuning could help minimize these fluctuations

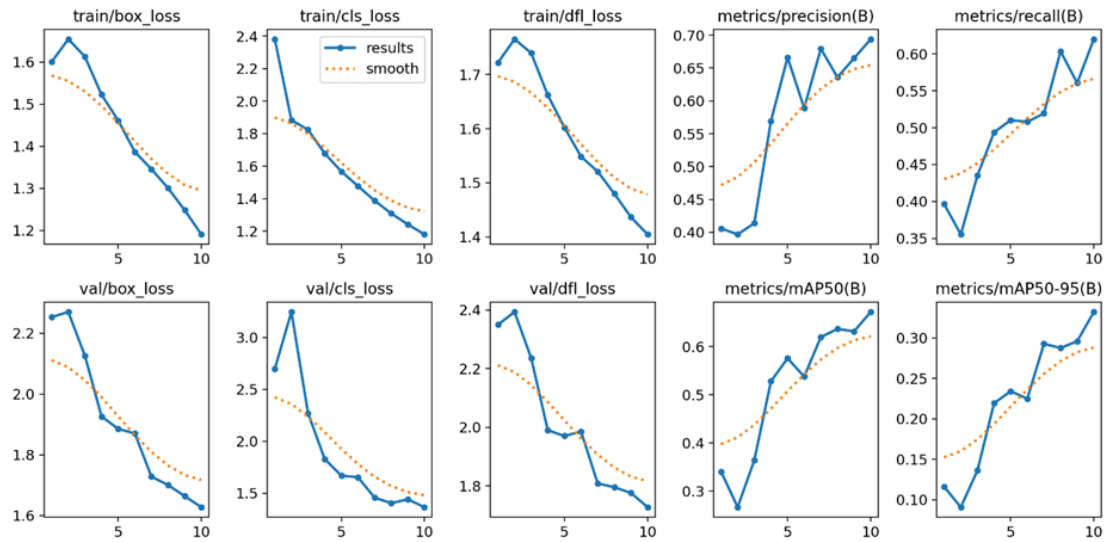


Fig 6: Training Loss Graph for YOLOv11s

Figure 6 shows the training loss curve for YOLOv11s shows a faster and more stable decline, indicating quicker convergence compared to YOLOv8s. This could imply that YOLOv11s is more efficient in terms of optimizing its parameters during training.

4.5 Predicted Images from both models

To visually assess the models' performance, we showcase some predicted images along with ground truth annotations. These images highlight how each model performs in detecting potholes under various conditions, including different lighting and angles.



Fig 7: Predicted Images from YOLOv8s



Fig 8: Predicted Images from YOLOv11s

Figure 7 and Figure 8 showcase the pothole detection results from YOLOv8s and YOLOv11s respectively. The YOLOv11s model outperforms YOLOv8s in multiple aspects including bounding box accuracy, average confidence scores, and robustness to visual noise. YOLOv11s achieves more consistent detection in the presence of shadows, reflections, and complex textures, with several high-confidence predictions (≥ 0.7). In contrast, YOLOv8s shows more fragmented and lower-confidence outputs. The comparative analysis highlights YOLOv11s as a more reliable model for daytime pothole detection tasks, demonstrating greater localization precision and reduced false positives.

5 DISCUSSION

This study conducted a comparative analysis of two advanced object detection models, YOLOv8s and YOLOv11s, for real-time pothole detection under diverse and challenging road conditions. The motivation was driven by the need for automated, accurate, and efficient road maintenance systems in regions like rural India, where traditional detection methods are labor-intensive and road conditions are challenging.

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv8s	0.7501	0.5786	0.6887	0.3314
YOLOv11s	0.6981	0.6267	0.6733	0.3320

Table 3: Comparison of key performance metrics for YOLOv8s and YOLOv11s

Results showed that YOLOv8s outperformed YOLOv11s in precision (0.7501) and mAP@0.5 (0.6887), making it more effective at accurately identifying potholes while minimizing false detections. It also achieved a reliable mAP@0.5:0.95 of 0.3314, indicating consistent performance across varying IoU thresholds. In contrast, YOLOv11s demonstrated higher recall (0.6267), meaning it was better at detecting all potholes, reducing the chances of missed detections a valuable trait for comprehensive road hazard systems. Although slightly lower in precision, YOLOv11s matched YOLOv8s in mAP@0.5:0.95 (0.3320) and surpassed it in inference speed and model efficiency, making it well-suited for real-time mobile or edge deployments with limited hardware resources.

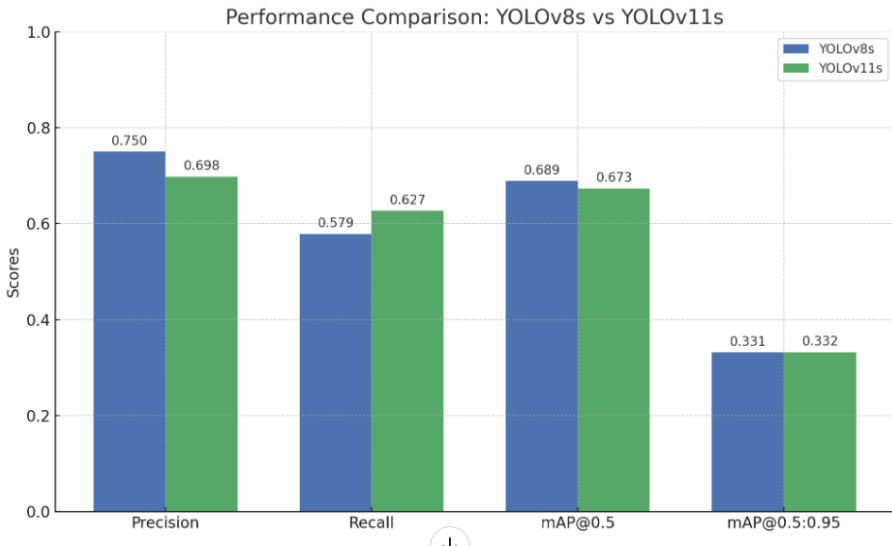


Fig 9: YOLOv8s & YOLOv11s Performance Comparison

Both models performed consistently well in challenging road scenarios, thanks to training on a diverse, contextually rich dataset. Based on these findings, YOLOv8s is recommended for applications where detection accuracy and precision are critical, such as automated reporting systems, while YOLOv11s is preferable for real-time detection on resource-constrained devices.

6 CONCLUSION

This study compared the performance of YOLOv8 and YOLOv11 models for pothole detection under diverse real-world conditions. YOLOv8s delivered higher accuracy and precision, making it ideal for applications requiring reliable road maintenance reporting. YOLOv11s offered faster inference speeds and greater efficiency, making it better suited for real-time detection on resource-limited devices. Both models maintained strong performance in challenging scenarios, such as nighttime and rural Indian roads with poor lighting and irregular surfaces. These results affirm the potential of YOLO-based deep learning approaches to enhance road safety through automated pothole detection systems.

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